

RQ Answers

omitted

2024-04-05

Data overview

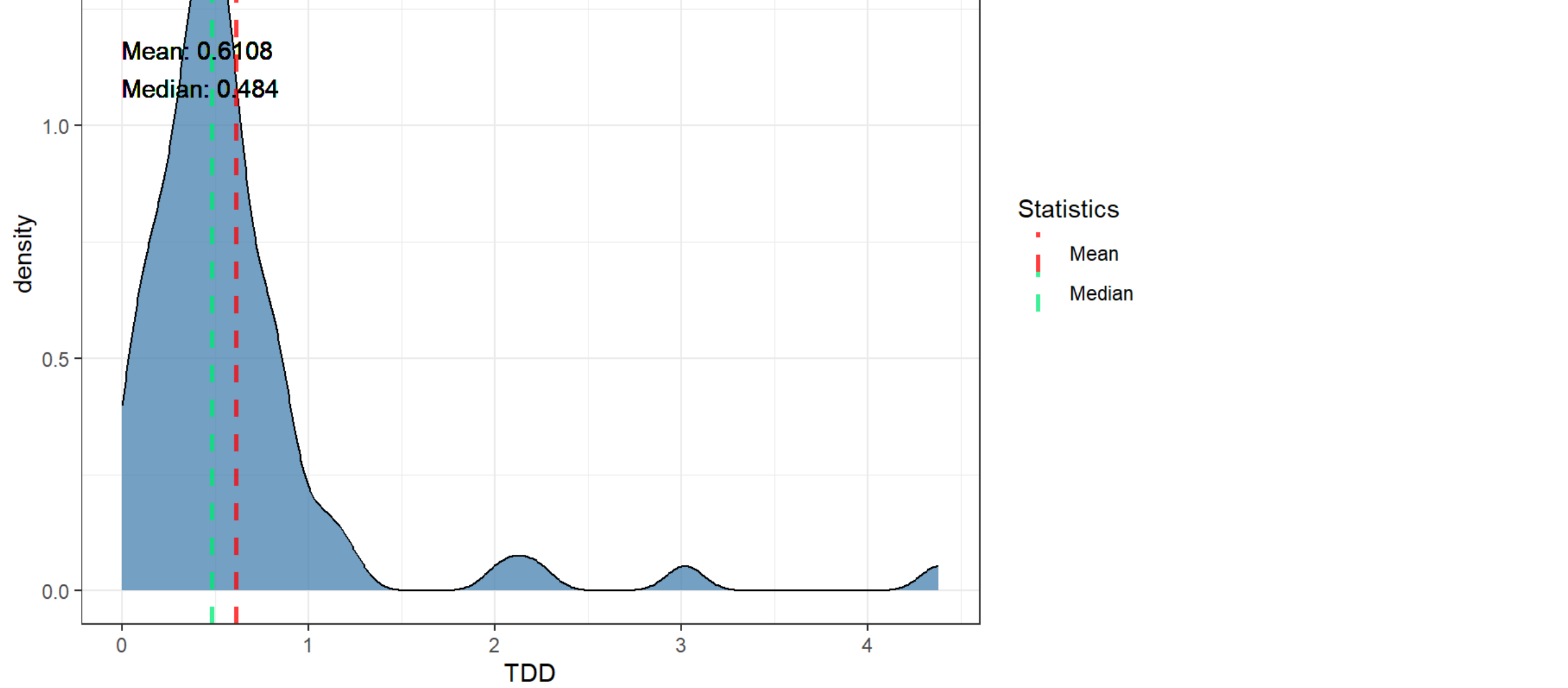
glimpse(all_data)				
## Rows: 9,121				
## Columns: 11				
## \$ repo	<chr> "opennlp", "opennlp", "opennlp", "opennlp", "opennlp", "opennlp", "opennlp", "opennlp", "opennlp", "opennlp", "opennlp"			
## \$ pr_number	<int> 102, 102, 102, 102, 102, 102, 102, 102, 102, 102, 122, ...			
## \$ rule	<chr> "java:S116", "java:S116", "java:S116", "java:S116", "java:S5993, ...			
## \$ severity	<chr> "MINOR", "MINOR", "MINOR", "MAJOR", "MAJOR", "MINO...			
## \$ file	<chr> "opennlp-tools/src/main/java/opennlp/tools/ml/maxe...			
## \$ type	<chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SML...			
## \$ status	<chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O...			
## \$ debt	<int> 2, 2, 2, 2, 5, 5, 5, 5, 10, 10, 270, 5, 10, 5, 10,...			
## \$ ncloc_affected_file	<int> 84, 84, 84, 84, 84, 84, 84, 84, 31, 106, 106, 106,...			
## \$ complexity	<int> 14, 14, 14, 14, 14, 14, 14, 14, 5, 11, 11, 11, 11,...			
## \$ origin	<chr> "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "P...			

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the data has 74 rows.
## i Please consider using `annotate()` or provide this layer with data containing a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but the data has 74 rows.
## i Please consider using `annotate()` or provide this layer with data containing a single row.
```

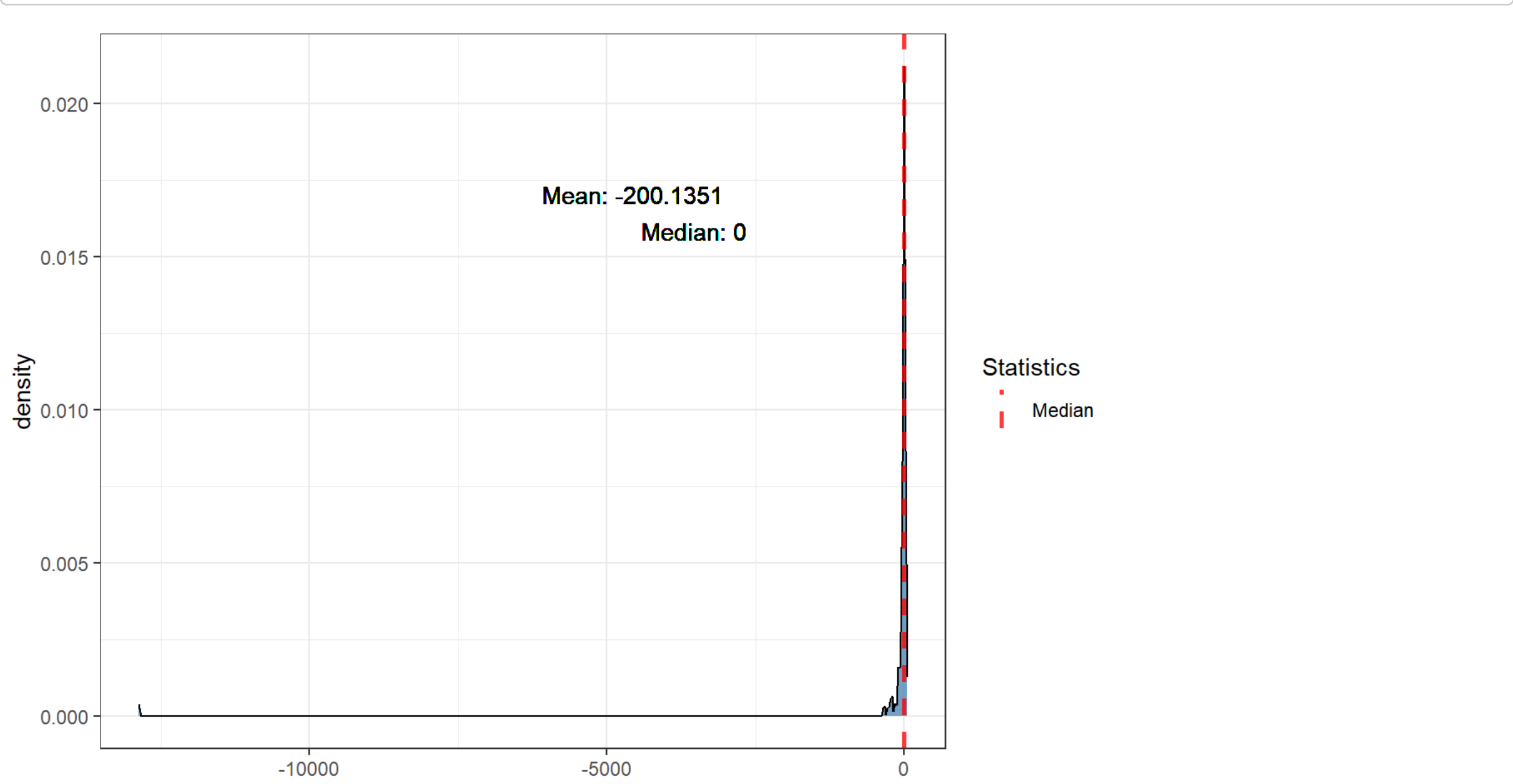


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the data has 74 rows.
## i Please consider using `annotate()` or provide this layer with data containing a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but the data has 74 rows.
## i Please consider using `annotate()` or provide this layer with data containing a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

## # A tibble: 1 × 4				
## repo	tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>	
## 1 opennlp	44.6	44.6	10.8	

Mean of percentages of TDV (mean of repo percentages)

## # A tibble: 1 × 3			
## mean_tdv_lower_than_zero	mean_tdv_equal_to_zero	mean_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>	
## 1	44.6	44.6	10.8

Percentages of pre-existing TDV by repo

## # A tibble: 1 × 3			
## repo	preexisting_tdv_equal_to_zero	preexisting_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 opennlp	42.5	57.5	

Mean of percentages of pre-existing TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## preexisting_tdv_equal_to_zero	mean_preexisting_tdv_greater_than_zero	
## <dbl>	<dbl>	
## 1	42.5	57.5

Percentages for new TDV by repo

## # A tibble: 1 × 3			
## # Groups: repo [1]			
## repo	new_tdv_equal_to_zero	new_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 opennlp	96.8	3.23	

Mean of percentages of new TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## mean_new_tdv_equal_to_zero	mean_new_tdv_greater_than_zero	
## <dbl>	<dbl>	
## 1	96.8	3.23

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank opennlp		
## <int> <chr>		
## 1	1 java:S1119	- 818
## 2	2 java:S117	- 555
## 3	3 java:S135	- 353
## 4	4 java:S1751	- 327
## 5	5 java:S1940	- 90
## 6	6 java:S1161	- 81
## 7	7 java:S1126	- 73
## 8	8 java:S3776	- 72
## 9	9 java:S1192	- 60
## 10	10 java:S1199	- 59

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S1119	122	
## 2 java:S117	123	
## 3 java:S135	124	
## 4 java:S1751	125	
## 5 java:S1940	126	
## 6 java:S1161	127	
## 7 java:S1126	128	
## 8 java:S3776	129	
## 9 java:S1192	130	
## 10 java:S1199	131	

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank opennlp		
## <int> <chr>		
## 1	1 java:S117	- 915
## 2	2 java:S1119	- 771
## 3	3 java:S1197	- 370
## 4	4 java:S100	- 342
## 5	5 java:S106	- 337
## 6	6 java:S1124	- 324
## 7	7 java:S3776	- 280
## 8	8 java:S125	- 200
## 9	9 java:S1192	- 194
## 10	10 java:S135	- 184

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S117	122	
## 2 java:S1119	123	
## 3 java:S1197	124	
## 4 java:S100	125	
## 5 java:S106	126	
## 6 java:S1124	127	
## 7 java:S3776	128	
## 8 java:S125	129	
## 9 java:S1192	130	
## 10 java:S1135	131	

Outliers

Higher TDD

## # A tibble: 74 × 5				
## repo	pr_number	total_debt	total_ncloc	tdd
## <chr>	<int>	<int>	<int>	<dbl>
## 1 opennlp	561	74742	17056	4.38
## 2 opennlp	122	320	106	3.02
## 3 opennlp	513	104	47	2.21
## 4 opennlp	545	732	358	2.04
## 5 opennlp	543	836	687	1.22
## 6 opennlp	474	7044	6308	1.12
## 7 opennlp	524	7440	6944	1.07
## 8 opennlp	218	110	113	0.973
## 9 opennlp	486	1461	1696	0.861
## 10 opennlp	214	271	318	0.852
## # i 64 more rows				

Higher TDV

## # A tibble: 74 × 3			
## pr_number	repo	tdv	
## <int> <chr>	<int>		
## 1	568 opennlp	47	
## 2	420 opennlp	40	
## 3	524 opennlp	34	
## 4	515 opennlp	20	
## 5	465 opennlp	15	
## 6	88 opennlp	7	
## 7	473 opennlp	5	
## 8	513 opennlp	5	
## 9	102 opennlp	0	
## 10	199 opennlp	0	
## # i 64 more rows			

Lower TDV

## # A tibble: 74 × 3			
## pr_number	repo	tdv	
## <int> <chr>	<int>		
## 1	561 opennlp	-12880	
## 2	474 opennlp	-3480	
## 3	543 opennlp	-263	
## 4	548 opennlp	-215	
## 5	547 opennlp	-210	
## 6	214 opennlp	-145	
## 7	545 opennlp	-102	
## 8	536 opennlp	-101	
## 9	449 opennlp	-95	
## 10	533 opennlp	-85	
## # i 64 more rows			

Extra

Number of PRs with pre-existing TD

## n	
## 1 73	

Number of java:S1192 issues that affect test code

## n	
## 1 0	

Number of java:S1192 issues that affect not test code

## n	
## 1 254	

Summary NCLOC

## ncloc	
## Min. :	26.0
## 1st Qu.:	127.8
## Median :	284.5
## Mean :	939.9
## 3rd Qu.:	584.8
## Max. :	16228.0